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GeoBased Online Marketplace for Local Markets
of
(BazaarLink)



Project Team:

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I.Executive Summary

BazaarLink is a web-based platform designed to digitally represent physical markets in Karachi. The system aims to bridge the gap between local vendors and customers through an interactive city-map interface. This report evaluates the technical, operational, and business feasibility of the project to ensure successful delivery within the semester timeframe.

II.Preliminary Requirements Analysis

Statement of Task

The task is to develop a centralized web platform that allows customers to explore Karachi's markets, browse specific shops, and place orders online, while providing shopkeepers with digital inventory management tools.

Preliminary Requirements

- **Functional:** User registration/authentication, interactive city-map for market browsing, shop and product management, order placement/tracking, and a review system.
- **Non-Functional:** Response times within 5 seconds, role-based access security, and a mobile-responsive UI.
- **Access Control:** Four distinct roles: Platform Admin, Market Admin, Shop Admin, and Customer.

Suggested Deliverables

- 1. Software Requirements Specification (SRS):** Detailed functional and non-functional requirements.
- 2. System Prototype:** A working web application featuring the interactive map and core ordering flow.
- 3. Database Schema:** A normalized SQL database design.

4. Final Report & Documentation: User guides and technical documentation for maintenance.

III.Process and Outline Plan

Methodology:

The team will follow an Iterative Development model. Unlike a linear approach, this model allows the team to build a "skeleton" of the BazaarLink platform early and then add complexity to the interactive map and role-based dashboards in successive cycles. This ensures that the core three-step navigation goal (Market → Shop → Product) is tested and refined multiple times before final delivery.

IV. Outline Plan and Milestones

The project is structured into five distinct phases over the semester:

Iteration	Activity	Timeline
Iteration 1	Feasibility Study and preliminary requirements completion.	Week 2
Iteration 2	Database design (SQL) and Backend architecture setup.	Week 5
Iteration 3	UI Development: Interactive City Map and Dashboards.	Week 8

Iteration 4	Integration of Order Management and Review systems.	Week 11
Iteration 5	Final testing, bug fixing, and project handover.	Week 14

V. Visibility and Communication Plan

To ensure the project stays aligned with client expectations, the following visibility measures will be implemented:

- **Client Contact:** Weekly progress updates via email and bi-weekly virtual demonstrations of iterative prototypes.
- **Internal Communication:** The team will use dedicated instant messaging (e.g., WhatsApp/Discord) for daily coordination and hold weekly in-person sync meetings.
- **Progress Tracking:** Use of a shared Kanban board (e.g., Trello or GitHub Projects) to visualize task status.

VI. Technical Requirements

The system will be built using a **three-tier architecture**:

- **Frontend:** HTML/CSS/JavaScript with a focus on a responsive, interactive map interface.
- **Application Layer:** Web framework (e.g., Node.js or Python) to handle business logic and role-based authentication.
- **Data Tier:** Relational SQL database (MySQL or PostgreSQL) for managing hierarchical data (City → Market → Shop).
- **Hardware:** Standard web servers and any device with a modern browser.

VII. Risk Analysis

Risk	Impact	Fallback Plan
Low Vendor Adoption	High	Start with a small pilot of 2-3 well-known markets to prove value.
Data Inconsistency	Medium	Implement strict validation rules for shopkeeper inputs.
Technical Complexity (Map)	Medium	Use established map APIs (e.g., Leaflet or Google Maps) instead of custom rendering.
Time Constraints	High	Prioritize core "Market-to-Order" flow; move AI recommendations to future enhancements.

VIII. Business Considerations

- **Scope Limitation:** Initial release excludes payment gateways and delivery logistics to simplify the business model.
- **Data Protection:** The system must comply with local e-commerce rules regarding the protection of customer data.
- **Scalability:** While initially limited to Karachi, the modular design allows for future expansion to other cities

VIII. CONCLUSION

Based on the feasibility analysis of the **BazaarLink** project, the team concludes that the system is technically and operationally viable within the 14-week semester timeframe. The project scope, centered on the core "Market → Shop → Product" navigation, is well-aligned with the team's current expertise in web development and SQL database management.

The platform effectively addresses the market gap in Karachi by digitally connecting physical vendors with local customers. While complex features like payment gateways and delivery logistics are excluded from the initial phase to ensure system stability, the modular three-tier architecture and normalized database design provide a scalable foundation for future enhancements.

Identified technical risks, including vendor adoption and data consistency, are proactively mitigated through a structured role-based access control system and a focused pilot rollout strategy. Therefore, the team is confident in its ability to fulfill all client requirements and recommends proceeding with the software development lifecycle for BazaarLink.